

HW Problem Set

Note: Meaning of a variable is same as we use in the class.

1. Two binary stars, A and B, have following epoch J2000.0 RA (α) and DEC (δ):

A :- $\alpha = 14^h 29^m 42.95^s, \delta = -62^\circ 40' 46.1''$

B :- $\alpha = 14^h 39^m 36.50^s, \delta = -60^\circ 50' 02.3''$

a) Find out the angular separation between A and B.

b) If the distance to A is 4×10^{16} m, then how far is A from B?

2. Show that the equation of motion of reduced mass μ for a binary system with total angular momentum L can be written as

$$r = \frac{L^2/\mu^2}{GM(1 + e\cos\theta)}.$$

Further comparing this EOM with polar equation of ellipse, one can find value of L for closed elliptic orbit with semimajor axis a as

$$L = \mu\sqrt{GMa(1 - e^2)}$$

3. Discuss how astronomers use Kepler's laws to determine individual star mass of a binary system when the plane of orbit has an inclination angle i w.r.t. plane of sky. During class, we discussed the case when $i = 0$.

4. Consider two stars in orbit about a mutual center of mass. If a_1 is the semimajor axis of the orbit of star of mass m_1 and a_2 is the semimajor axis of the orbit of star of mass m_2 , prove that the semimajor axis of the orbit of the reduced mass is given by $a = a_1 + a_2$.

5. Do problem 7.4 of Carroll and Ostlie.

6. Sun's apparent magnitude is -26.83. What will be its absolute magnitude? A star at a distance 4 pc has an apparent magnitude 2. What will be its absolute magnitude? Given that the Sun has a luminosity 3.9×10^{26} W, find the luminosity of the distant star using above information.